

MAJOR PROJECT - I | BACKEND DEVELOPMENT PROGRESS REPORT

Blockchain-Assisted Distributed AES Key Lifecycle Management Framework for Secure Healthcare Internet of Things

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1. Progress Summary

Following project enrollment and finalization of the system architecture (Review 1), backend development began on the Healthcare Key Management module. As of this report, four foundational backend phases have been completed, tested, and pushed to GitHub. The team is now proceeding to Phase 5, with the target of having a working prototype ready before the Mid-Term Evaluation (4th Week of September 2026).

Overall Status: 4 / 14 backend phases complete (Phases 1–4) | Next Milestone: **Phase 5 — Key Generation & Secure Key Storage**

2. Completed Backend Phases

Phase	Module	Status	Key Work Completed
Phase 1	Project Foundation	● Completed	Spring Boot project setup, MySQL connection, JPA/Hibernate configuration, Maven build, base project structure.
Phase 2	Key Management Foundation	● Completed	Key entity and metadata model, key repository/service layer, initial key APIs, key lifecycle fields in schema.
Phase 3	Authentication & Authorization	● Completed	User + Role model, registration, BCrypt password hashing, JWT-based login, JWT filter, Spring Security config, protected /api/keys/** endpoints.
Phase 4	AES-256 Encryption	● Completed	AES-256-GCM implementation with random IV generation, encryption/decryption service, verified with unit tests.

Verification: Login flow tested end-to-end — returns 200 OK with a valid Bearer token. All Phase 1–4 code has been committed and pushed to the team's GitHub repository.

3. Current Position — Backend Roadmap (14 Phases)

The diagram below shows the full backend roadmap for the Healthcare Key Management System, with completed, in-progress, and upcoming phases:



4. Next Milestone — Phase 5

Phase 5 — Key Generation & Secure Key Storage

This phase implements secure AES key generation, safe handling of key material, key metadata storage, and key versioning — building directly on the key-management foundation from Phase 2. Target: working prototype ready before the Mid-Term Evaluation (4th Week of September 2026).

Upcoming Phases (6–14) — Brief Overview

Phase	Focus Area
Phase 6	Key Lifecycle Management — Active, Rotation, Expiry, Revocation, versioning & timestamps.
Phase 7	Secure Key Retrieval — JWT + role + permission checks before any key access, with audit.
Phase 8	Role-Based Access Control — granular Admin / User permission enforcement.
Phase 9	Audit Logging — who, what, which key, when, and result for every security operation.
Phase 10	Exception Handling & API Hardening — global exception handler, validation, proper HTTP codes.
Phase 11	API Documentation — Swagger / OpenAPI integration.
Phase 12	Testing — unit, integration, security, repository, and controller tests.